

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0318

Fairplay 138 kV
August 23, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0318. The Interconnection Customer proposes a 160.1 MW wind facility interconnecting at the Fairplay 138 kV substation in the Pennsylvania area. The request was received on March 19, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 189 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,324,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tuco – John Day 138kV L	State Campus – Kiya 138kV line	159.5	231	5.1
Matanzas – John Day 138kV	Kiya – State Campus 138kV	156.8	240	5.1
Prides Corner 138-1 kV Auto	American River – State Campus 138kV	145.3	438	9.9
Aburn Rd 500/345 kV XFMR	Texas Eastern 5 – Acec Dellwood 345kV	132.5	676	27.2
Prides Corner 345/1kV Auto	American Riv – State Campus 138kV L	130.4	488	9.9
Tuco – Medora 138kV L	Burwell – Youngwood 500kV Line	110.3	206	6.7
Lutesville – Sunrise Ctr 345kV	Cayucos – Sunrise Ctr 500kV	109.3	1,202	6.9
Pride Corner – Pit 7 Tap 138kV	Prides Corner Substation – Morgantown Energy Facility 345kV Line	108.6	250	6.7
W Campus – Edes Tap 2 345kV	ShawvilleStation – Orchard Rd 500kV	106.0	1,155	6.2
Wahiawa – N Allegheny Windpower Project 345kV	Sunrise Center – Merret 500kV	104.7	1,447	6.9
Ft Humbug – Orchard Road 500kV L	West Campus – Ft Humbug 345kV	104.1	3,014	10.3
Edes Tap 2 – Lutesville 345kV Line	Orchard Road – Shawville Station 500kV	103.7	1,191	5.9
W Campus – Fort Humbug 345kV	Orchard Road – Fort Humbug 500kV	102.9	2,193	7.7
LittleChute – Talenenergy Martins Creek Llc Harrisburg 345kV line	Coronet – Ft Humbug 500kV L	102.3	1,626	6.5
Texas Eastern 5 345-1 kV Autotransformer	Texas Eastern 5 – Ronco 500kV L	101.8	563	18.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Youngwood – Univ Of Nc Chapel Hill Cogen Facility 345kV	Youngwood – Texas Eastern 5 500kV	100.8	1,387	22.3
Auburn Rd – Prides Corner 345kV line	Auburn Road 345/138kV Transformer Bank	100.8	1,244	7.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tuco – John Day 138kV L	Rebuild line with higher-capacity conductor	\$1,582,000	2% (\$32,000)
Matanzas – John Day 138kV	Rebuild line with higher-capacity conductor	\$1,817,000	2% (\$37,000)
Prides Corner 138/1 kV Transformer	Replace transformer with higher-rated unit	\$3,945,000	2% (\$90,000)
AuburnRoad 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$28,615,000	11% (\$3,093,000)
Prides Corner 345/1kV Autotransformer	Replace transformer with higher-rated unit	\$21,156,000	2% (\$481,000)
Tuco – Medora 138kV Line	Rebuild line with higher-capacity conductor	\$1,097,000	4% (\$40,000)
Sunrise Ctr – Lutesville 345kV L	Reconductor line	\$15,543,000	3% (\$448,000)
Pit 7 Tap – Prides Corner 138kV	Reconductor line	\$1,944,000	12% (\$240,000)
Edes Tap 2 – W Campus 345kV	Reconductor line	\$10,804,000	2% (\$261,000)
Wahiawa – N Allegheny Windpower Project 345kV	Reconductor line	\$7,539,000	3% (\$233,000)
Orchard Rd – Ft Humbug 500kV Line	Reconductor line	\$19,325,000	2% (\$332,000)
Edes Tap 2 – Lutesville 345kV L	Reconductor line	\$19,177,000	2% (\$475,000)
West Campus – Ft Humbug 345kV	Reconductor line	\$31,572,000	2% (\$531,000)
LittleChute – Talenenergy Martins Creek Llc Harrisburg 345kV line	Reconductor line	\$8,476,000	2% (\$150,000)
Texas Eastern 5 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$21,490,000	9% (\$1,983,000)
Youngwood – Univ Of Nc Chapel Hill Cogen Facility 345kV	Reconductor line	\$13,062,000	6% (\$785,000)
Auburn Rd – Prides Corner 345kV line	Reconductor line	\$5,149,000	2% (\$113,000)
	Total allocated to project		\$9,324,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.